Research Exposé

MQTT for MicroServices

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1. Topic

This project investigates the use of MQTT (Message Queuing Telemetry Transport) — a lightweight, publish/subscribe messaging protocol originally designed for constrained IoT environments — as a communication mechanism between microservices, in comparison to the conventional REST/HTTP-based approach.

A small microservices system will be implemented in two functionally equivalent versions: one using synchronous REST calls for inter-service communication, and one using asynchronous, broker-based MQTT messaging. The two versions will then be evaluated and compared under a range of controlled conditions.

2. Motivation

Microservice architectures conventionally rely on REST/HTTP for inter-service communication. This creates tight coupling between services: if a downstream service is temporarily unavailable or slow, the calling service's request fails immediately, and both services must be online and reachable at the same time.

MQTT decouples communication through a broker: publishers and subscribers do not need to be available simultaneously, and messages can be queued until a receiver is ready. This model is well established in IoT, where networks are often unreliable and devices are resource-constrained, but it is comparatively underexplored as a general-purpose communication pattern for backend microservices outside the IoT domain.

As distributed systems continue to grow in scale and are increasingly expected to remain resilient to partial failures, understanding whether MQTT's decoupling and resilience advantages translate into measurable benefits for general microservice communication is of practical relevance for system design decisions.

3. Research Questions

1. How does MQTT-based service communication compare to REST-based communication in end-to-end latency, under both normal and degraded network conditions?
2. How resilient is an MQTT-based microservice architecture to a service crashing or becoming temporarily unavailable, compared to a REST-based architecture handling the same scenario?
3. What are the differences in resource usage (CPU and memory) between an MQTT-broker-based architecture and a REST-based architecture at comparable scale?
4. How does the choice of MQTT Quality of Service (QoS) level affect the trade-off between message delivery reliability and performance overhead?

4. Planned Approach

A small system of three to four services (e.g. a sensor/data simulator, a processing service, and an alerting/notification service) will be implemented in two versions — one communicating via REST, one via MQTT using a broker such as Eclipse Mosquitto. Both versions will be containerized to ensure a comparable deployment environment.

Controlled experiments will then be run against both versions, including simulated service outages, injected network latency, and increasing load, while measuring latency, message delivery success, and resource consumption.

5. Tentative Timeline

- Weeks 1–2: Literature review, environment setup, implementation of the REST-based version
- Weeks 3–4: Implementation of the MQTT-based version
- Weeks 5–6: Experiment design, execution, and data collection
- Final weeks: Analysis of results and write-up